

The Arduino™’s integrated development environment (IDE a.k.a. its software) simplifies the above tasks by providing a ‘library’, which contains the commonly used functions to simplify the software part of programming the microcontroller. At the hardware end, the system clock, programming headers, and push-down and pull-up resistors are taken care of on the board itself. All we need to do is plug the hardware to the sensors or other devices and start working without learning too many details about the architecture or worrying about components necessary to run the microcontroller.

Those among you well aware of microcontroller programming will appreciate the simplicity of the Arduino™ environment. Since it’s an open source project, you can freely access the schematics required to build your Arduino™ compatible boards. If you plan on using the ATmega microcontroller for a project, you can just use the Arduino™ programming environment to program the microcontroller and make minor modifications to the existing hardware.

Hardware and Software

Arduino™ creators have created many different variants of Arduino™ to cater to different areas of usage. Most Arduino™ boards have the Atmel ATmega series based processor as their microcontroller. Others have the ARM processor based on SMART series and can be identified by the word ‘SAM’ in their name. Check out this list of all available Arduino™ boards. Once you’re familiar with the products, we’ll guide you in selecting the right board to match your need.

As you can see in the list, except for the Arduino™ Due, all other processors are based on an ATmega series microcontroller from Atmel.

Each piece of hardware in the list is capable enough to perform functions such as controlling motors, locating your current position using GPS, building game pads with tilt and play functionality, _____

(insert your idea here)
etc. The list of applications is endless.



Selecting an Arduino™ board

Selecting the one for your creation...

With the wide range of Arduino™ products available in market,